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Steering systems, steering and speed coordination systems, and associated vehicles

Abstract

In a broad respect, vehicles that are capable of making a low- to zero-radius turn using the independent rotation of drive wheels and by turning the non-driving steerable structure or structures (such as wheels) with a steering input device (in some embodiments, the driving wheels also may be capable of being turned). This may be accomplished using a steering system, a speed control system and an integration device (together, a control system) that are configured to work together to provide correct steering in forward and reverse, and, in some embodiments, to reduce the speed of the outboard drive wheel of the vehicle when it enters an extreme turn under constant speed input. Different systems configured for use in such vehicles are included.

Inventors: Schaedler; Axel (Olmsted Falls, OH), Hauser; Hans (Strongsville, OH), Ruebusch; Rick (El Paso, TX), Cornwell; Ian David (Lancashire, GB)

Applicant: MTD Products Inc (Valley City, OH)

Family ID: 37683826

Assignee: MTD PRODUCTS INC (Valley City, OH)

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Primary Examiner: Neacsu; Valentin

Assistant Examiner: Stabley; Michael R

Attorney, Agent or Firm: Norton Rose Fulbright US LLP

Background/Summary

CROSS-REFERENCE(S) TO RELATED APPLICATION(S) (1) This is a continuation of co-pending U.S. patent application Ser. No. 14/664,457, filed Mar. 20, 2015, which is a continuation of U.S. patent application Ser. No. 13/081,482, filed Apr. 6, 2011, now U.S. Pat. No. 9,409,596, which is a divisional of U.S. patent application Ser. No. 11/490,881, filed Jul. 21, 2006, now U.S. Pat. No. 7,992,659, which claims priority to U.S. Provisional Patent Application Ser. No. 60/701,716, filed Jul. 22, 2005, U.S. Provisional Patent Application Ser. No. 60/710,231, filed Aug. 22, 2005, and U.S. Provisional Patent Application Ser. No. 60/731,593, filed on Oct. 26, 2005, the contents of all of which are incorporated by reference.

BACKGROUND

1. Field of the Invention

(1) The invention relates generally to vehicles that have low to zero turning radius capability. Zero turning radius vehicles are often described as ZTR vehicles. However, this name has also been used to described vehicles capable of a turning radius that is not precisely zero. More specifically, the invention relates to steering systems, steering and speed coordination systems, and vehicles that comprise one or both types of systems.

2. Description of Related Art

(2) ZTR vehicles are generally propelled by rear drive wheels, which can be driven at different speeds to accomplish steering. The speed and direction of rotation of the drive wheels of some ZTR vehicles are controlled through separate hand levers. Some users find these levers confusing because they control both vehicle speed and direction.

(3) Some ZTR vehicles use a steering wheel instead of separate control levers. However, some of these vehicles do not provide correct steering when the vehicle is in reverse. For example, when backing up and turning the steering wheel to make a left-hand turn, some of these vehicles produce

a right-hand rear turn where the front of the vehicle—instead of the rear—moves to the left. See U.S. Pat. No. RE 34,057 as an example of such a ZTR vehicle.

(4) John Deere introduced a series of Spin-Steer Technology™ (SST) tractors. The SST tractors possess a rear-wheel driven differential steering system controlled by a steering wheel, and a vacuum-actuated reverse logic system that provides for conventional steering in reverse. The front wheels are caster wheels that are not steerable. See U.S. Pat. No. 6,257,357 for a description of these tractors.

(5) U.S. Pat. No. 6,601,663 discloses a ZTR vehicle that utilizes a steering wheel to control steering, and a single hydraulic variable displacement pump and dual variable displacement hydraulic motors, each of which is coupled to a ground engaging wheel that is used to steer and drive the vehicle. This ZTR vehicle provides for proper steering in the forward and reverse directions.

(6) U.S. Patent Application Publication No. 2003/0102171 also discloses a ZTR vehicle capable of proper wheel-effected steering in forward and reverse. The independently-actuated rear wheels drive the vehicle. They also steer the vehicle by rotating at different speeds and/or directions.

(7) One problem with using caster wheels as non-steerable front wheels on ZTR vehicles is noticeable when driving on the side of a hill. Gravity will tend to pull the vehicle down the hill. This may cause the portion of the vehicle supported by the caster wheels to turn downhill against the operator's wishes. Additionally, when attempting to turn the ZTR vehicle uphill, the drive wheels may lose traction as the operator tries to produce the torque required to get the castor wheels pointed in the uphill direction.

(8) Steerable front wheels have been used on ZTR vehicles. See U.S. Pat. No. 3,362,493 (Davis, et al.) and U.S. Pat. No. 5,042,238 and U.S. Patent Application Publication No. 2003/0019682. However, each has shortcomings. For example, the Davis patent device is not equipped with a system that can reduce the speed of the outboard drive wheel of a vehicle entering an extreme turn at a constant speed input.

(9) U.S. Pat. Nos. 6,196,342 and 6,129,164 disclose reverse steering logic mechanisms that are coupled to and interact with a dual differential type of drive and steer transmission to cause the transmission to execute vehicle turns in the direction that the steering wheel is turned when operating in forward or reverse. These patents disclose the use of caster wheels, and do not disclose the use of steerable front wheels.

(10) U.S. Pat. No. 6,921,109 discloses a reverse steering logic mechanism and a mechanism for providing “variable steering responsiveness.” It discloses using these mechanisms with the dual differential type transmission in U.S. Pat. No. 6,196,342.

(11) U.S. Pat. No. 6,905,985 discloses a complicated system of linkages that purportedly provides for steering control of front steerable wheels and transmission-effected steering that rotates the rear wheels such that the vehicle turns in the direction that the steering wheel is turned when operating in forward or reverse. This patent discloses the use of this system in combination with a dual differential type transmission.

(12) U.S. Pat. No. 6,152,248 discloses the use of a non-circular gear pair in the steering of a vehicle, but that gear pair does not control the turning of a non-driving wheel.

SUMMARY

(13) In a broad respect, the invention relates to vehicles that are capable of making a low- to zero-radius turn (e.g., a small radius turn) using the independent rotation of drive wheels and by turning the non-driving wheel or wheels with a steering input device (in some embodiments, the driving wheels may also be capable of being turned). This may be accomplished using a steering system, a speed control system and an integration device (together, a control system) that are configured to work together to provide correct steering in forward and reverse, and, in some embodiments, to reduce the speed of the vehicle (specifically the outboard drive wheel) when it enters a sufficiently extreme turn (e.g., one in which the ground engaging wheel can be turned no further) under

constant speed input.

(14) In some embodiments, these vehicles comprise a frame; a steerable structure (such as a ground-engaging wheel, which also may be characterized as a non-driving wheel) coupled to the frame; two drive wheels coupled to the frame; a transmission system capable of driving the two drive wheels at different speeds and in different directions; a steering assembly configured to control the steerable structure; a speed control assembly coupled to the transmission system; and an integration device that integrates a steering input with a speed input to steer and drive the vehicle. The steering assembly, the speed control assembly and the integration device are configured to work together to reduce the speed of the outboard drive wheel during an extreme turn while the speed input received by the speed control assembly is constant.

(15) In some embodiments, these vehicles comprise a frame; a steerable structure (such as a ground-engaging wheel) coupled to the frame; two drive wheels coupled to the frame; a transmission system capable of driving the two drive wheels at different speeds and in different directions; a steering assembly configured to control the steerable structure; a speed control assembly coupled to the transmission system, the speed control assembly including a speed input device configured to be manipulated by an operator; and an integration device that integrates a steering input with a speed input to produce a blended output for steering and driving the vehicle that is transmitted to the transmission system as a result of an operator manipulating the speed input device. The steering assembly, the speed control assembly and the integration device are configured to work together to steer the vehicle correctly in both forward and reverse during a turn. Stated another way, the steering assembly, the speed control assembly and the integration device are configured to work together such when the vehicle is turned, the direction of the turn is the same for a given steering input whether the vehicle is traveling in forward or reverse. As a result, the direction of the turn does not change when going from forward to reverse.

(16) In another respect, the invention relates to a driving and steering system that comprises at least one steering cam configured to receive a steering input and be coupled to and articulate a non-driving wheel; a speed cam coupled to the steering cam and movable in response to a speed input; and an assembly coupling the steering cam to the speed cam. The system can include two steering cams positioned on opposite sides of a steering input device (such as a steering wheel), and a speed cam can be coupled to each of the speed cams to form two pairs of steering and speed cams. The steering cams can be configured to have the same shape, and the speed cams can be configured to have the same shape. The assembly can be configured to move the steering cams in opposite directions in response to a given steering input and to move the speed cams in the same direction in response to a given speed input.

(17) In another respect, the invention relates to a driving and steering system that comprises two steering cams that move in opposite directions in response to a steering input; a speed cam coupled to each steering cam and movable in response to a speed input; and an assembly coupling each steering cam to one of the speed cams.

(18) In another respect, the invention relates to a steering system that comprises a first gear pair that controls the turning of a non-driving wheel (meaning that the transmission system is not involved with such control), the first gear pair including a non-circular drive gear that engages a non-circular driven gear. Each gear pair of the system can be designed to cause the non-driving wheels to follow a vehicle turn radius that matches (or at least substantially matches) the vehicle turn radius produced by the driving wheels (under the control of the transmission system).

(19) In another respect, the invention relates to a steering system that comprises a gear pair having a non-uniform gear ratio, the gear pair being configured to control the turning of a non-driving wheel (meaning that the transmission system is not involved with such control).

(20) In another respect, the invention relates to a vehicle that comprises a frame; at least two non-driving wheels coupled to the frame; at least two drive wheels coupled to the frame; a transmission system capable of (a) driving the two drive wheels at different speeds and in different directions

and (b) causing the drive wheels to produce a first vehicle turning radius; and a steering assembly configured to cause the non-driving wheels to produce a second vehicle turning radius, the steering assembly including two pairs of non-circular gears configured such that the second vehicle turn radius can be equal to the first vehicle turn radius during operation of the vehicle.

(21) In another respect, the invention relates to a steering system in a vehicle having at least two non-driving wheels, at least two drive wheels, and a transmission system capable of (a) driving the drive wheels at different speeds and in different directions and (b) causing the drive wheels to produce a first vehicle turn radius, the steering system comprising: first and second pairs of non-circular gears configured to work together to cause the non-driving wheels to produce a second vehicle turning radius that is equal to the first vehicle turning radius for a given steering input.

(22) In another respect, the invention relates to a steering system in a vehicle having at least two non-driving wheels, at least two drive wheels, and a transmission system capable of (a) driving the drive wheels at different speeds and in different directions and (b) causing the drive wheels to produce a first vehicle turn radius, the steering system comprising: first and second pairs of gears that each have a non-uniform gear ratio and that are configured to work together to cause the non-driving wheels to produce a second vehicle turning radius that is equal to the first vehicle turning radius for a given steering input.

(23) In another respect, the invention relates to a steering system that comprises a first gear pair including a first drive gear coupled to a first driven gear that is coupled to a king pin, the first gear pair being configured to rotate the king pin through a greater angle in response to an inward turn caused by a first steering input than in response to an outward turn caused by a second steering input that is equal in magnitude but opposite in direction to the first steering input.

(24) Different aspects of these devices (e.g., vehicles) and systems, as well as other devices and systems, are described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following drawings illustrate by way of example and not limitation. Identical reference numerals do not necessarily indicate an identical structure. Rather, the same reference numeral may be used to indicate a similar feature or a feature with similar functionality. Every feature of each embodiment is not always labeled in every figure in which that embodiment appears, in order to keep the figures clear. At least FIGS. 5-13 are drawn to scale, meaning the sizes of the depicted elements are accurate relative to each other for at least one set of embodiments of the present devices and systems.

(2) FIG. 1 is a perspective view of a lawn and garden type vehicle;

(3) FIG. 2A is a top view of the steering assembly and front axle of the vehicle of FIG. 1;

(4) FIG. 2B is a top view of the speed control assembly and the transmission system of the vehicle of FIG. 1;

(5) FIGS. 3A and 3B schematically illustrate the positions of the steerable, ground-engaging front wheels of an embodiment of the present vehicles;

(6) FIG. 4 is a partial perspective view of the steering and speed control assemblies of the vehicle of FIG. 1 coupled together with an integration device;

(7) FIG. 5 illustrates a perspective view of the front axle of the vehicle of FIG. 1;

(8) FIG. 6A is an enlarged partial perspective view of one of the front wheel assemblies of the vehicle of FIG. 1;

(9) FIGS. 6B-6E are enlarged partial perspective views of different embodiments of front wheel assemblies that may be used with the vehicle of FIG. 1;

(10) FIG. 7 illustrates a perspective view of another embodiment of the front axle of the vehicle of

FIG. 1;

(11) FIGS. 8A-8C illustrate views of a gear pair used with the front wheel assembly of FIG. 6;

(12) FIG. 9A-9C illustrate views of an alternate embodiment of a gear pair used with the front wheel assembly of FIG. 7;

(13) FIG. 10 is a perspective view showing aspects of the of the speed control assembly of FIG. 2B;

(14) FIG. 11 is a perspective view showing the interaction between the steering assembly and the speed control assembly of the vehicle of FIG. 1;

(15) FIG. 12 is a close-up view of one of the present steering control members in the form of a steering cam;

(16) FIG. 13 is a close-up view of one of the present speed control members in the form of a speed cam;

(17) FIGS. 14A-14C show the position of the speed control member from FIG. 13 in neutral, forward and reverse, where the vehicle is steered straight ahead;

(18) FIGS. 15A-15C show the position of the speed control member from FIG. 13 in neutral, forward and reverse, where the vehicle is in a maximum turn and the depicted speed control member is on the inboard side of the turn;

(19) FIG. 16 charts the speed of the wheels for one embodiment of the present vehicle versus the applied steering for a constant speed input;

(20) FIG. 17 is a top view of an alternate embodiment of a steering assembly, a speed control assembly, and an integration device that may be used with the vehicle of FIG. 1;

(21) FIGS. 18 and 19 are different perspective views of a variable pitch worm of the steering assembly of FIG. 17;

(22) FIG. 20 is a side view of a portion of the arrangement shown in FIG. 17;

(23) FIG. 21 is an exploded view of another embodiment of a steering assembly, a speed control assembly, and an integration device that may be used with the vehicle of FIG. 1;

(24) FIG. 22 is a perspective view from below of the system of FIG. 21;

(25) FIG. 23 is a perspective view from above of the system of FIG. 21;

(26) FIGS. 24 and 25A-25D represent, in schematic form, various configurations of the system of FIG. 21;

(27) FIG. 26 is a cross-sectional illustration of an embodiment of a steering and speed control assembly;

(28) FIG. 27 is a plan view of a further embodiment of portions of the system of FIG. 21;

(29) FIG. 28 is a side view of the embodiment of FIG. 27;

(30) FIG. 29 is a section in a longitudinal plane through a transmission suitable for use as one of the present drive units;

(31) FIG. 30 is a schematic representation of the transmission of FIG. 29; and

(32) FIG. 31 is a cross-sectional view (without the cross hatching) of the transmission of FIG. 29 looking in the direction of arrows III-III.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(33) The terms “comprise” (and any form of comprise, such as “comprises” and “comprising”), “have” (and any form of have, such as “has” and “having”), “contain” (and any form of contain, such as “contains” and “containing”), and “include” (and any form of include, such as “includes” and “including”) are open-ended linking verbs. Thus, a vehicle that “comprises” a frame; a steerable structure coupled to the frame; two drive wheels coupled to the frame; a transmission system capable of driving the two drive wheels at different speeds and in different directions; a steering assembly configured to the steerable structure; a speed control assembly coupled to the transmission system; and an integration device that integrates a steering input received by the steering assembly with a speed input received by the speed control assembly to steer and drive the vehicle; where the steering assembly, the speed control assembly and the integration device are

configured to work together to reduce the speed of the outboard drive wheel during an extreme turn while the speed input received by the speed control assembly is constant, is a vehicle that possesses the listed elements, but is not prohibited from possessing elements that are not listed (such as an additional steerable structure).

(34) Likewise, an element of an apparatus that “comprises,” “has,” “contains” or “includes” one or more features possesses those one or more features, but is not limited to possessing only those one or more features. Furthermore, a structure that is configured in a certain way must be configured in at least that way, but also may be configured in a way or ways that are not specified.

(35) The terms “a” and “an” are defined as one or more than one unless this disclosure explicitly requires otherwise. The terms “substantially” and “about” are defined as at least close to (and includes) a given value or state (preferably within 10% of, more preferably within 1% of, and most preferably within 0.1% of).

(36) General Configuration

(37) Referring now to the figures, FIG. 1 illustrates a vehicle **10**, such as a lawn and garden tractor. The vehicle **10** includes a prime mover **12**, such as an engine, that is mounted to a structural frame or chassis **14**. The vehicle **10** includes drive wheels **16**, such as left and right rear drive wheels that are coupled to the frame **14**. The drive wheels **16** are operatively coupled to the engine **12** through a transmission system to provide locomotion to the vehicle **10**. The vehicle **10** also has steerable structure **18**, such as right and left front ground-engaging wheels, which may be non-driving wheels. Other embodiments of the vehicles have only one steerable structure (e.g., three-wheeled all-terrain vehicles). Furthermore, in some embodiments, steerable structures such as skis may be used instead of wheels.

(38) The chassis **14** supports an operator station comprising a seat **22**. Vehicle **10** also includes a mower deck **26** mounted to the vehicle **10** in any manner chosen with sound engineering judgment. The invention is applicable to other types of vehicles, including but not limited to utility vehicles, off road vehicles, tractors, golf carts, and even automobiles.

(39) As shown in FIGS. 2A and 2B, the front wheels **18** are coupled to the frame of the vehicle through a pivotable connection to a front axle **19** mounted on the chassis **14**. The front wheels **18** are also coupled to a steering assembly **20**, which is configured to control the direction they turn as discussed more fully below. In the embodiment of the present vehicles shown in the figures, the front wheels are the steerable wheels **18** and the rear wheels are the drive wheels **16**. However, one skilled in the art will understand that the rear wheels may be the steerable wheels and the front wheels may be the drive wheels without departing from the scope of the invention. Likewise, the front wheels may be both the steerable wheels and the drive wheels.

(40) A steering input device **24** (which is part of the embodiment of the steering assembly **20** shown in the figures) and a speed input device **28** (which is part of the embodiment of the speed control assembly discussed below) are located near the seat **22** (FIG. 1) so that they are accessible to the operator of the vehicle. An operator may apply a steering input to the steering input device **24**, which transfers the steering input to the steering assembly **20**. Steering input device **24** may take the form of a conventional steering wheel. However, the steering input device **24** may be another suitable steering device, including, but not limited to, a steering rod or joystick (not shown).

(41) The speed input device **28** provides a speed input to the balance of the speed control assembly **21**, and (at least in part) regulates the forward and reverse speed of the vehicle **10**. Speed input device **28** may take the form of a single pedal, such as a treadle pedal arrangement mounted on a single shaft. In such an embodiment, the speed input device **28** is rocked forward to select forward drive, or rocked backward to select reverse drive. The speed input device **28** may be biased toward a central position that corresponds to a neutral or stationary condition.

(42) Vehicle **10** also includes an integration device **27** that is configured to integrate a steering input received by the steering assembly **20** via the steering input device **24** with a speed input received

by the speed control assembly (discussed below) via the speed input device **28** to drive and steer the vehicle **10**. The configurations of the present steering assemblies, speed control assemblies and integration devices allow the vehicle to make small- to zero-radius turns.

(43) The left and right drive wheels **16** are driven through a transmission system that, in the depicted embodiment, comprises left and right drive units **29**. Vehicle **10** includes a speed control assembly **21** that controls the direction and magnitude of rotation of the rear drive wheels **16**. The drive units **29** may be transmissions of the continuously variable type, capable of providing a continuous range of ratios from forward to reverse. Examples of a suitable transmission utilizing a ratio varying-device, or variator, in conjunction with an epicyclic shunt gear to provide a geared neutral facility is described in International Application PCT/GB03/00332, published under WO 03/064892, and International Application PCT/GB03/02332, published under WO 03/100295, both of which are incorporated by reference for those descriptions. Alternately, the drive units **29** may be hydrostatic transmissions (HST) or electric motors, both of which are well known in the art. The drive units **29** may be used to independently drive the drive wheels **16**.

(44) The driver dictates the speed and direction of the vehicle **10** by manipulating the steering input device **24** and the speed input device **28**, which transmit the steering and speed inputs received from the driver to the balance of the steering and speed control assemblies that are linked by the integration device **27**. The manner in which the steering and speed control assemblies work together through the integration device to drive and steer the vehicle is described in more detail below. In the embodiment of vehicle **10** shown in the figures, the amount of torque that the rear drive wheels must produce to turn the vehicle **10** is reduced because front wheels **18** are steerable. In contrast, the drive wheels **16** of some conventional ZTR vehicles with non-steerable castor wheels must produce significant torque to cause the castor wheels to react and point in the desired direction. Furthermore, a certain amount of familiarity and skill is required to prevent skidding the inboard drive wheel and tearing the grass under the wheel.

(45) In the embodiment of vehicle **10** shown in the figures, the right and left drive wheels **16** are coupled to chassis **14** such that their direction is fixed and their rotational axes are in constant alignment. In contrast, the front steerable wheels **18** are coupled to the chassis **14** in a way that gives them the ability to change direction. FIGS. 3A and 3B are schematic top views of the vehicle **10** illustrating that it possesses the ability to achieve substantially true Ackermann steering. FIG. 3A shows a non-zero radius turn, and FIG. 3B shows a zero-radius turn. When front wheels **18** make the turn depicted in FIG. 3A, they take two distinct arc-like paths $P_{sub.i}$ and $P_{sub.0}$, which ideally will have a common center point C located along the axis that extends through the center of both drive wheels **16**. Lines $L_{sub.i}$ and $L_{sub.0}$ extend from center point C and intersect the paths $P_{sub.i}$ and $P_{sub.0}$, respectively, of the two wheels at the rotational centers of the wheels. The use of a substantially-true Ackermann steering geometry (which can be achieved using some of the embodiments discussed below) can help to avoid scrubbing rubber from the tire tread on the outboard wheel or damaging vegetation under the front wheels.

(46) Steering Assembly **20**

(47) Aspects of steering assembly **20** are depicted in, e.g., FIGS. 2A-12. One function of the steering assembly **20** is to couple the steering input device **24** to the front steerable wheels **18** to aid in guiding vehicle **10**. Another function of the steering assembly **20** is to provide a steering input to the integration device **27**, which can coordinate that steering input with a speed input received through the speed input device **28**. Another function of the steering assembly **20** is its ability to turn the vehicle **10**, even in a zero turning radius mode (or a small turning radius mode), while receiving an input from a conventional steering input device such as a steering wheel.

(48) In one embodiment, the steering assembly **20** includes a steering shaft **30** extending downwardly from the steering input device **24** and terminating in a toothed steering pinion **32**. The steering shaft **30** is rotatably coupled to the chassis **14** with a bushing **34** or any other suitable means using sound engineering judgment. The steering shaft **30** and pinion **32** take the steering

input received through the steering input device **24** and take part in transmitting it to front wheel assemblies **50**, which then convert the steering input into desired steering angles of the front wheels **18**, as explained below. In one embodiment, the coupling between the steering shaft **30** and the front wheel assemblies is accomplished using, in part, left and right bevel gears **36**. The pinion **32** is positioned between and simultaneously engages the left and right bevel gears **36** such that rotation of the pinion **32** causes simultaneous rotation of the left and right bevel gears **36**. The steering input device **24** and steering pinion **32** may be rotated through about 120 degrees of movement. For example, the steering input device **24** may be selectively rotated 60 degrees in a first direction with respect to a neutral steering position and 60 degrees in a second direction. However, the steering input device **24** and steering pinion **32** may be configured for rotation through any range of angles suited to a given application.

(49) Rotating the steering input device **24** and pinion **32** in a first direction causes one of the bevel gears **36** to rotate forward or toward the front of the vehicle **10** and the other bevel gear **36** to rotate backward or toward the rear of the vehicle **10**. The left and right bevel gears **36** are coupled to left and right jack shafts **38**, respectively. Preferably, the left and right sides of the steering assembly **20** are substantially identical but mirror images of each other. Accordingly, only the right side of the steering assembly **20** will be described below.

(50) As shown FIG. **4**, the jack shaft **38** is positioned generally orthogonal to the steering shaft **30** and is coupled to a steering mechanism **40** at its outer end. In one embodiment, the steering mechanism is a steering cam **40**. The steering cam **40** is coupled to the jack shaft **38** so that it may be rotated by movement of the jack shaft **38** in first and second directions about pivot **41**, through which the axis of the jack shaft **38** extends. An outer portion of the steering cam **40** is coupled to a drag link **42**. In the FIG. **4** embodiment, when the steering cam **40** is rotated in a clockwise direction (which would occur during an outboard turn for the depicted steering cam), the drag link **42** moves forward or toward the front of the vehicle **10**, and when the steering cam **40** is rotated in a counter-clockwise direction (which would occur during an inboard turn), the drag link **42** moves toward the rear of the vehicle **10**. (The direction the drag link **42** moves depends on the position of the drag link **42** with respect to the pivot **41**.) Thus, rotation of the steering input device **24** is transmitted into forward or aft movement of the drag link **42**. Preferably, the drag link **42** is coupled to the steering cam **40** with a suitable linkage **44**, such as a ball linkage. The drag link **42** is also coupled to a front wheel assembly **50**, which converts the steering input received by the steering input device **24** into a steering angle of the front wheel **18**. More specifically, the front wheel assembly **50** translates the position of the support structure about which front wheel **18** rotates in response to the steering input received through the drag link **42** from the steering input device **24**.

(51) Steering input device **24** may be coupled to front wheel assemblies **50** in other ways in other embodiments using sound engineering judgment.

(52) Turning to FIG. **5**, the front wheel assembly **50** includes a steering or driving gear **52** pivotably mounted on a post **54** received in the front axle **19**. The steering gear **52** has a linking portion **56** to which the drag link **42** may be coupled with a suitable connector, such as a ball connector **58**. As best seen in the enlarged view of FIG. **6A**, the steering gear **52** has teeth **60**, one or more of which mesh with one or more of the teeth **62** of wheel or driven gear **70**. The wheel gear **70** is coupled to the front wheel **18** in order to steer the front wheel to the left or right. In one embodiment, the wheel gear **70** is mounted on a king pin **74** so that rotation of the wheel gear **70** causes rotation of the king pin **74**. In the illustrated embodiment, the king pin **74** has a square head **75** about which the wheel gear **70** rotates. The king pin **74** is pivotably coupled to the chassis **14** of the vehicle **10** by virtue of being rotatably mounted to the front axle **19** using a suitable bearing, bushing or the like **79**. A pivot shaft **76** extends generally orthogonal to the king pin **74**, and the front wheel **18** is rotatably mounted on the pivot shaft **76**.

(53) As FIG. **6A** shows, the steering gear **52** can pivot about post **54** due to the force transmitted

through the drag link **42** as a result of the steering input received through the steering input device **24**. The rotation of steering gear **52** is transmitted to the wheel gear **70** to change the direction of the front wheel **18**. The front wheel assemblies **50** enable the two front wheels **18** to be driven in substantially true Ackermann steering geometry.

(54) In one embodiment, the linking portion **56** to which the drag link **42** is coupled is positioned inward of the post **54** about which the steering gear **52** pivots and to the rear of a line L connecting the two posts **54**, as best seen in FIG. 2A. Line L is generally parallel to the transverse axis of the vehicle **10** and perpendicular to the longitudinal or major axis of the vehicle **10**. During a turn, the drag link **42** on the inboard side moves in a first direction (e.g., to the rear) while the drag link **42** on the outboard side of the turn moves in a second direction (e.g., to the front). Movement of the inboard drag link **42** causes the linking portion **56** to move further to the rear with respect to the post **42** and away from line L as the steering gear **52** pivots around the post **54**. On the outboard side, the outboard drag link **42** moves forward, causing the linking portion **56** to move forward toward the front of the vehicle as the steering gear **52** pivots.

(55) At first, the outboard linking portion **56** moves closer to the line L. Continued rotation of the steering input device **24** may cause the linking portion **56** to pass through the line L and then move away from, and forward of, line L. Steering assembly **20**, and more specifically each wheel assembly **50**, is configured such that the magnitude of the component of the movement of the drag link **42** that causes rotation of the steering gear **52** increases as the linking portion **56** moves away from line L. Thus, the movement of the drag link **42** on the inboard side in the rear direction causes a larger rotational movement of the steering gear **52** on the inboard side than the forward movement of the drag link **42** on the outboard side. Therefore, the inboard front wheel **18** rotates faster and further to contribute to the substantially true Ackermann steering geometry.

(56) As shown in FIGS. 5 and 6A, the front axle **19** is preferably not straight. Instead, it has non-linear portions **90** near either end that have a forward slanting portion **91** joined to a rear slanting portion **92**. Each rear slanting portion **92** leads to an outer portion **93** of the axle **19** located near where the front wheel **18** is mounted. Each non-linear portion **90** of the front axle **19** forms a pocket **94** to the rear of the front axle **19** that receives the front steerable wheel **18** on the inward side during an extreme turn. The pocket **94** allows the inward front steerable wheel **18** to be turned greater than 90 degrees, and preferably between 100 and 120 degrees as illustrated in FIG. 3B, without having the rear portion of the front wheel **18** on the inside of the turn contact the front axle **19**.

(57) Other gear arrangements besides those shown in FIGS. 5 and 6A may be used for front wheel assemblies **50**. For example, FIGS. 7 and 9A-9C, discussed below, show other non-circular gears that may be used for front wheel assemblies **50**. Some additional alternatives are shown in FIGS. 6B-6E. These figures depict an enlarged partial view of the front right wheel assembly **50**. In contrast to FIG. 6A, the pocket **94** of front axle **19** in each of FIGS. 6B-6E is facing the viewer of the figure, and the drag link **42** and ball connector **58** are swiveled away from the viewer (such as to be used with a steering pinion oriented in front of the front axle **19**), though if used with the version of steering assembly **20** shown in FIG. 2A would point back toward the viewer. FIG. 6B shows an example of a front wheel assembly **50** comprising a driving gear **52** and a wheel gear **70** that are both circular and coupled with a chain **59**. The post **54** on which driving gear **52** is mounted extends up through lever **51**. The lever **51** may be coupled to the driving gear **52** using any suitable means, such that rotation of the lever **51** about the axis of the post **54** also causes a rotation of the steering gear **52**. The angle of the lever **51** with respect to the “straight ahead” position of the driving gear **52** can be set (taking into account other relevant factors, such as the coupling between the driving and driven gears, and the manner in which the front steering assemblies are coupled to each other) to provide substantially true Ackermann steering (as is true of embodiments shown in FIGS. 6C and 6D).

(58) FIG. 6C shows an example of a front wheel assembly **50** comprising a driving gear **52** and a

wheel gear **70** that are both circular and coupled with a belt **59A**.

(59) FIG. **6D** shows an example of a front wheel assembly **50** comprising a driving gear **52** and a wheel gear **70** that are both circular. The two gears are coupled by virtue of one or more of the teeth **60** of the driving gear meshing with one or more of the teeth **62** of the driven gear.

(60) FIG. **6E** shows another embodiment of front wheel assembly **50**. Lever **51** is coupled to the planet carrier **53** of planetary gear **57**. Planet carrier **53** is coupled to the king pin, which controls the articulation of the pivot shaft **76**. The ring **71** of the planetary gear **57** is coupled through an arm to the post **54**, which is shorter in this embodiment and does not extend through to the bottom of the front axle **19**. The angle of the lever **51** with respect to the “straight ahead” position of the planet carrier **53** can be set (taking into account other relevant factors, such as the orientation of the planet carrier relative to the king pin, and the manner in which the front steering assemblies are coupled to each other) to provide substantially true Ackermann steering.

(61) FIG. **7** illustrates another embodiment of the wheel assembly **50**. In this embodiment, the linking portion **56** is even with the post **54** about which the steering gear **52** pivots when in the neutral position so that the linking portion **56** and the post **54** are aligned parallel to the transverse axis of the vehicle **10**. In this embodiment, the drag links **42** on either side of the vehicle move in opposite directions but cause the same magnitude of rotation of the two steering gears **52**. In this embodiment, the shape of the gears **52** and **70** cause the inboard front wheel **18** to rotate faster and further to provide the desired Ackermann steering geometry because they are configured as shown in FIGS. **9A-9C** and described below. Preferably, a front tie bar **78** couples the two wheel assemblies **50** together to provide structural support. Such a tie bar can be used to couple the two wheel assemblies shown in the FIGS. **5** and **6A-6E** embodiments as well.

(62) One purpose of the front tie bar is to aid in distributing loads, such as when one of the front wheels **18** hits a curb or other object. The force from striking the object can be distributed through both wheel assemblies **50** through the front tie bar and then to the chassis **14**. This reduces the shock that is transmitted back through the steering system to the steering input device **24** and felt by the operator.

(63) Non-Circular Gears

(64) Turning now to FIGS. **8A-8C**, in one embodiment, the steering gear **52** and the wheel gear **70** combine to form a non-circular gear pair **81**. In one preferred embodiment, the steering gear **52** has a shape comprising two spline portions **82**, **84** connected by a valley portion **86**. As seen in FIG. **8A**, the distance from pivot axis A.sub.s of the steering gear **52** to the pitch line P.sub.s of the steering gear **52** in the spline portions **82**, **84** is greater than the distance from the pivot axis A.sub.s of the steering gear **52** to the pitch line P.sub.s of the steering gear **52** in the valley portion **86**. The rear portion **85** of the steering gear **52** can have any shape selected to accomplished the desired steering, such as the shape depicted in FIG. **6A**. The wheel gear **70** has a substantially parabolic shaped portion **87** having a vertex **88**. The rear portion **89** of the wheel gear **70** can have any shape selected to accomplish the desired steering, such as the shape depicted in FIGS. **8A-8C**.

(65) In the neutral or straight-ahead position, at least one or more of the teeth **62** near the vertex **88** of the parabolic portion **87** of the wheel gear **70** engage at least one or more of the teeth **60** in the valley portion **86** of the steering gear **52** as illustrated in FIG. **8A**. As the steering gear **52** is rotated around its axis A.sub.s, one of the spline portions **82**, **84** engages the side of the parabolic portion **87** as the driven wheel gear **70** rotates around its axis A.sub.w, as illustrated in FIGS. **8B** and **8C**.

(66) In one embodiment, the spline portions **82**, **84** of the steering gear have a different number of teeth. In the illustrated embodiment, the spline portion **82** has five teeth **60** and the spline portion **84** has seven teeth **60**. The spline portion **84** has additional teeth **60** that extend further around the steering gear **52** on the side that engages the wheel gear **70** during an inward turn. The inward front wheel **18** must turn through a greater angle than the outboard front wheel **18** to meet the Ackermann geometry. Accordingly, the spline portion **82** that engages the wheel gear **70** when making a turn on the outward side does not need as many teeth **60** because the outward front wheel

18 does not turn as far.

(67) The non-circular shapes of the steering gear **52** and the wheel gear **70** (and, more specifically, the non-circular shapes of the toothed portions of the steering and wheel gears) enable the gear combination to have a non-uniform gear ratio. In the neutral position, the ratio of the distance between the pivot axis A.sub.s of the steering gear **52** to the pitch line P.sub.s of the steering gear **52** to the distance between the pivot axis A.sub.w of the wheel gear **70** and the pitch line P.sub.w of the wheel curve is preferably between about 1.0:1.0 and 2.0:1.0, and more preferably about 1.5:1.0. In the extreme turning position illustrated in FIG. **8C**, the ratio of the distance between the pivot axis A.sub.s of the steering gear **52** to the pitch line P.sub.s of the steering gear **52** to the distance between the pivot axis A.sub.w of the wheel gear **70** and the pitch line P.sub.w of the wheel curve is preferably between about 2.0:1.0 and 4.0:1.0, and more preferably about 3.0:1.0. However, any gear ratio suited to the application may be chosen. Thus, in a preferred embodiment, the output of the gear ratio may range from 1.0:1.0 to 4.0:1.0, and more preferably from 1.5:1.0 to 3.0:1.0 as the gears rotate as shown in FIGS. **8A**, **8B** and **8C**.

(68) The position of linking portion **56** on drive gear **52** and the non-uniform gear ratio of the gear pair permits the steering angle of the front wheels **18** to be responsive to the magnitude of the desired turn as determined by the input to the steering input device **24**. When the vehicle **10** is traveling straight ahead or in a slight turn and the steering input device **24** is close to the neutral position, it is preferable for the movement of the steering input device **24** to cause only relatively small changes in the angle of the front wheels **18**. This enables the operator to travel in straight lines and precisely control the vehicle. On the other hand, when the operator desires to perform an extreme turn, it is useful for the movement of the steering input device **24** to cause a relatively larger corresponding change in the steering angle of the front wheels **18**. Accordingly, in some embodiments, the steering system **20** is configured such that movement of the steering input device **24** in the plus or minus twenty degree range from neutral causes a relatively small change in the steering angle of the vehicle. However, when the steering input device **24** is turned for an extreme turn, such as a zero radius turn, the steering assembly **20** increases the change in the steering angle so that the front wheels **18** rapidly reach the larger steering angle.

(69) For example, some embodiments of the steering assembly **20** may be configured such that movement of the steering input device **24** to a position between about 10 degrees and about 20 degrees from the neutral position causes a corresponding change of the steering angle of the vehicle of between about 5 and about 20 degrees. In such embodiments, movement of the steering wheel to a position between about 20 degrees and about 40 degrees from neutral causes a corresponding change of the vehicle steering angle of between about 20 and about 60 degrees. In such embodiments, movement of the steering wheel to a position between about 40 degrees and about 60 degrees from neutral causes a corresponding change of the steering angle of between about 60 and about 120 degrees. Dimensions of the steering and wheel gears of a given gear pair, such as the pitch lines, may be set so that the rotational axes of both front steerable wheels **18** are always made to intersect with the single point C on the rotational axis of drive wheels **16** to provide substantially true Ackermann steering.

(70) FIGS. **9A-9C** illustrate another embodiment of a non-circular gear pair **81A**. This gear pair **81A** has non-uniform pitch lines such that the shapes of the steering gear **52A** and wheel gear **70A** produce substantially true Ackermann steering geometry. This gear pair **81A** may be used with the embodiment of the wheel assembly **50** shown in FIG. **7**.

(71) The steering gear **52A** has a shape comprising two spline portions **82A**, **84A** connected at a juncture **86A**. The spline portion **82A** is engaged when the front wheel **18** to which the gear pair **81A** is coupled is on the outboard side of the turn and the spline portion **84A** is engaged when the front wheel **18** is on the inboard side of the turn. In the FIG. **9A** embodiment, the distance from pivot axis A.sub.s of the steering gear **52A** to the pitch line P.sub.s of the steering gear **52A** in the spline portion **82A** is substantially constant throughout the spline portion **82A**, such that this

portion of the steering gear **52A** resembles a sector of a circle. However, the distance from pivot axis A.sub.s of the steering gear **52A** to the pitch line P.sub.s is non-uniform in the spline portion **84A**. Accordingly, the embodiment of steering gear **52A** may be characterized as a non-circular gear, or as having a non-circular toothed portion.

(72) Preferably, the distance from pivot axis A.sub.s to pitch line P.sub.s progressively increases to between about 110% and about 150% of the distance to the pitch line at the juncture **86A**. In the illustrated embodiment, the distance from pivot axis A.sub.s to the pitch line P.sub.s near the teeth that engage the wheel gear **72A** during an extreme inward turn is about 123% of the pitch line at the neutral position. The rear portion **85A** of the steering gear **52A** can have any suitable shape, such as the shape shown in FIG. 7.

(73) The wheel gear **70A** also has a non-uniform pitch line configured to match the pitch line of the steering gear **52A**. In the illustrated embodiment, the wheel gear **72A** has a first portion **83A** in which the distance from the pivot axis A.sub.w of the wheel gear **70A** to the pitch line P.sub.w of the wheel gear **70A** is substantially constant throughout the portion **83A**, such that this portion of the wheel gear **70A** resembles a sector of a circle. The wheel gear **70A** has a non-uniform portion **87A** in which the distance from the pivot axis A.sub.w of the wheel gear **70A** to the pitch line P.sub.w of the wheel gear **70A** in the portion **87A** is non-uniform. The uniform and non-uniform portions meet at a juncture **88A**.

(74) In the neutral or straight-ahead position, one or more of the teeth **62A** near the juncture **88A** of the wheel gear **70A** engage one or more of the teeth **60A** near the junction **86A** of the steering gear **52A** as illustrated in FIG. 9A. When making an inward turn as illustrated in FIGS. 9B and 9C, the steering gear **52A** is rotated around the axis A.sub.s such that the spline portion **84A** engages the non-uniform side **87A** of the wheel gear **70A** as the wheel gear **70A** rotates around axis A.sub.w.

(75) Preferably, the distance from pivot axis A.sub.w to the pitch line P.sub.w progressively decreases to between about 50% and about 75% of the distance at the juncture **88**. In the illustrated embodiment, the distance from pivot axis A.sub.s to the pitch line P.sub.s near the teeth that engage the wheel gear **72A** during an extreme inward turn as shown in FIG. 9C is about 65% of the pitch line at the neutral position. The rear portion **89A** of the wheel gear **70A** can have any shape selected using sound engineering judgment, such as the shape shown in FIG. 7.

(76) In one embodiment, the position of the teeth **60A**, **62A** and the pitch lines P.sub.s and P.sub.w for the steering gear **52A** and wheel gear **70A** are chosen so that substantially true Ackermann steering is provided by the gear pair **81A**. One method of selecting the pitch lines P.sub.s and P.sub.w begins with determining the desired steering angles for the inside and outside front wheels **18**. Referring back to FIG. 3A, the inside wheel steering angle α and outside wheel steering angle ω can be determined using the following formula:

$$\tan(90^\circ - \omega) = [\tan(90^\circ - \alpha) - L + W] / L \quad [\text{Equation 1}]$$

(77) Using the desired steering angles, the pitch lines P.sub.s and P.sub.w may be set so that the rotational axes of both front steerable wheels **18** are always made to intersect with a single point C located on the rotational axis of drive wheels **16**, as seen in FIGS. 3A and 3B.

(78) In the illustrated embodiment, the portions of the steering gear **52A** and wheel gear **70A** that engage each other when the gears are on the outside position of a turn (spline portion **82A** and portion **83A**) have uniform pitch lines, while the portions of the gears that engage each other when the gears are on the inside position of the turn (spline portion **84A** and portion **87A**) have non-uniform pitch lines. However, all portions of the gears can be non-uniform as long as the pitch lines P.sub.s and P.sub.w are selected to produce a substantially true Ackermann steering geometry for turning the front wheels **18**.

(79) The front wheel **18** on the inboard side of a turn steers through a greater steering angle than the outboard front wheel **18** in order to meet the Ackermann geometry. However, in the embodiment of the gear pair shown in FIGS. 9A-9C, the steering gears **52A** on the inboard and outboard sides of the vehicle **10** will be rotated by the steering system at substantially the same

speed and substantially the same magnitude. Preferably, the steering gear **52A** is configured to rotate about 90 degrees, with about 45 degrees in the spline portion **82A** and about 45 degrees in the spline portion **84A**. The spline portion **84A** has a longer pitch line than the spline portion **82A**, and therefore more teeth. In the illustrated embodiment, the spline portion **82A** has six teeth **60A** and the spline **84A** has seven teeth **60A**. Similarly, the portion **87A** of the wheel gear **70A** must match its corresponding spline portion **84A** on the steering gear **52A**, so it also has a greater number of teeth **62A**. As the pitch line P.sub.w gets closer to the axis A.sub.w in the portion **87A**, the teeth **62A** extend a greater distance around the circumference of the wheel gear **70A**. As a result, the gear teeth **62A** in the portion **83A** take up a sector of between about 70 and 89 degrees and the gear teeth **62A** in the portion **87A** take up a sector of between about 91 and 120 degrees. The variation in the pitch lines between the inward turn side (**84A**, **87A**) and the outward turn side (**82A**, **83A**) causes the inward front wheel **18** to achieve a greater steering angle than the outward front wheel **18** in accordance with the Ackermann steering geometry.

(80) The non-circular shapes of the steering gear **52A** and the wheel gear **70A** enable the gear combination to have a non-uniform gear ratio. In the neutral position, the ratio of the distance between the pivot axis A.sub.s and pitch line P.sub.s of the steering gear **52A** to the distance between the pivot axis A.sub.w and pitch line P.sub.w of the wheel gear **70A** is preferably between about 1.0:1.0 and 2.0:1.0, and more preferably about 1.5:1.0. The spline portion **82A** of the steering gear **52A** and the portion **83A** of the wheel gear **70A** have uniform pitch lines; therefore this ratio remains substantially constant for the front wheel **18** on the outboard side of the turn. However, in the extreme turning position illustrated in FIG. **9C**, the ratio of the distance between the pivot axis A.sub.s and the pitch line P.sub.s of the steering gear **52A** to the distance between the pivot axis A.sub.w and the pitch line P.sub.w of the wheel gear **70A** for the front wheel on the inboard side is preferably between about 2.0:1.0 and 4.0:1.0, and more preferably about 3.0:1.0. However, any ratio suited to a given application may be chosen.

(81) Steering and Speed Control Assemblies with the Integration Device

(82) Referring back to FIGS. **2B** and **4**, the speed control assembly shown generally at **21** and its interaction with the steering assembly **20** via the integration device **27** to control the transmission drive units **29** will now be described. In a preferred embodiment, the integration device **27** includes components that mechanically integrate a steering input from the steering assembly **20** corresponding to the position of the steering input device **24** with a speed input corresponding to the position of the speed input device **28** to drive and steer vehicle **10**. The integration device **27** that is shown in the figures is configured to set the direction of rotation of each drive wheel **16** in response to the steering input the integration device receives from the steering assembly **20**. The integration device, steering assembly and speed control assembly depicted in, for example, FIGS. **1-16** are configured to work together to reduce the speed of (such as by decelerating) the outboard drive wheel of the vehicle in a sufficiently extreme turn, even when the speed input is constant (see FIG. **16**). In some other embodiments, the steering and speed control assemblies and the integration device are not configured in that manner.

(83) The integration device **27** includes an assembly **101**, such as a linkage assembly, that couples the speed control assembly **21** and steering assembly **20** to the transmission drive units **29** such that the steering and speed inputs can be coordinated to control the magnitude and direction of rotation of the transmission drive units **29**.

(84) In one embodiment, the assembly **101** includes pintle links **102** pivotably coupled to the transmission drive units **29**. When the pintle links **102** are pivoted in first and second directions, they provide input to the transmission drive units **29** to control the direction and magnitude of the rotational output of the transmission drive units, and thus the direction and magnitude of rotation of the respective drive wheels **16**. The more that the pintle links **102** are pivoted, the greater the magnitude of speed at which the drive units **29** are driven in each respective direction. The assembly **101** may also include drive rods **104**, which may be pivotably coupled to the pintle links

102 at distal ends **105** of the drive rods. The drive rods **104** are movable back and forth so as to pivot the pintle links **102** in the first and second directions. The drive rods **104** may be independently shifted with respect to the other. "Independently shifted" means that the drive rods **104** may be moved separately, such as in the longitudinal direction of the vehicle. As a result, the pintle links **102** are independently pivoted such that the transmission drive units **29** can drive their associated drive wheels at different rates and in different directions, although they may also drive them at the same rate and in the same direction. Drive rods **104** may be configured in any suitable fashion to accommodate the orientation of the transmission system (and, more specifically, the transmission drive units). For example, two sections of a drive rod (or two drive rods) may be coupled together longitudinally using complimentary bell cranks (see FIG. **21** for an example of one bell crank) or a connecting plate (see FIGS. **10** and **11**). Alternatively, a change in the height of drive rod may be accomplished by bending it (see FIG. **4**).

(85) The speed control assembly **21** of vehicle **10** includes a speed input shaft **110** that is coupled to the chassis **14** in a way that allows it to rotate in response to movement of the speed input device **28** to which it is coupled (e.g., through a fixed attachment). Speed input device **28** is coupled to speed input shaft **110** such that the speed input shaft **110** will rotate in the same general direction that the speed input device **28** is depressed. When the steering input device **24** is in a neutral position (not steered to the left or right), rotating the shaft **110** in either direction will cause the left and right drive units **29** to drive at substantially the same magnitude and in the same direction, propelling the vehicle **10** straight forward or backward. The speed input device **28** may be biased via a spring or other mechanism toward a neutral or non-driving position.

(86) As shown in FIG. **10**, the speed input shaft **110** is coupled to a speed mechanism **112**. The speed mechanism comprises two speed cams **112**, one controlling the left drive unit **29** and the other controlling the right drive unit **29**. The speed input shaft **110** is coupled to an arm **113** with a bracket **114**. The arm **113** is coupled to a second speed shaft **115** through bracket **116**. Thus, the speed input shaft **110** is coupled to the second speed shaft **115** through the arm **113** such that rotation of the speed input shaft **110** is transmitted into rotation (in the same direction) of the second speed shaft **115**.

(87) Each speed cam **112** is coupled to the second speed shaft **115** preferably with a bracket **117** at point **125**. Each speed cam **112** has a speed slot **119**. Integration device **27**, and more specifically linkage assembly **101**, includes a follower **120** that is coupled to the end of the drive rod **104** and rides in the speed slot **119**. In the illustrated embodiment, for example, the follower **120** includes a yoke **121** having a pin **122** configured to ride in the speed slot **119**. The follower **120** may contain rollers, bearings or other components to enable the follower **120** to slide in the speed slot **119**.

(88) As FIG. **10** shows, actuation of the speed input device **28** applies rotational force equally to both of the speed cams **112**. The speed cams **112** rotate about a pivot point **118** positioned on a line extending along the axis of the second speed shaft **115** and located within the speed slot **119** (see FIG. **13**) as a result of the configuration of brackets **117**, which act as bridges. The speed slot **119** is preferably curved so that the follower **120** can freely slide from one end of the speed slot **119** to the other as the drive rod **104** is pivoted about a center axis positioned near the pintle link **102**.

Furthermore, speed slot **119** may be shaped like an arc having a radius that is equal to the distance from the pivot point **118** to the actuation location, which is the location where the pintle links control actuation of the drive units. As a result, a speed input that cause the follower **120** to move in the speed slot will not actuate either of the drive units because the distance between the pintle link and a line defining the arc of the speed slot (which runs through the pivot point **118**) is constant all along the slot.

(89) FIG. **11** shows the follower **120** received in the speed slot **119** of the speed cam **112** at a bottom position. This may be the default or biased position. However, the neutral position may be at the top of the speed slot **119** depending on the arrangement of the drive rod **104** and the pintle link **102** and how the pintle link **102** is configured to control the drive units **29**. The speed control

assembly **21** receives the steering input from the steering assembly **20** via the two steering cams **40**. Each steering cam **40** is coupled to the speed cam **112** with a steering command arm **124**. The steering command arm **124** has a generally V-shaped body and is coupled to the chassis **14** at pivot **126**. One end of the steering command arm **124** contains a follower link **128** that is movably coupled to the steering cam **40**. Specifically, in this embodiment, the steering cam **40** has a steering slot **127** that receives the follower link **128**. The other end of the steering command arm **124** is coupled to the drive rod **104** with a slide **133**. The slide **133** may be pinned to the steering command arm **124** in any suitable manner (see FIG. 4) such that it can pivot about its pinned axis and translate along the length of a portion of the drive rod **104** without disrupting the longitudinal position of the drive rod and actuating one of the drive units **29**. The steering command arm **124** can selectively move the follower **120** in and along the length of the speed slot **119**. As a result, the position of the steering cam **40** can control the position in the speed slot **119** where the follower **120** engages the speed cam **112**.

(90) As shown in FIG. 12, the steering slot **127** on the steering cam **40** has a dwell portion **130** that has a first contour for controlling the position of the follower link **128** when the steering cam **40** is on the outboard side of the vehicle **10** during a turn. The dwell portion **130** may include an end section **130A** that has a different contour than an inner section **130B** of the dwell portion **130**. The steering slot **127** also has a cam portion **131** that has a second contour for controlling the position of the follower link **128** when the steering cam **40** is on the inboard side of the vehicle **10** during a turn. The first contour of the dwell portion **130** is different from the second contour of the cam portion **131**. The cam portion **131** may have an end section **131A** and an inner section **131B**. The end section **131A** may have a different contour than the an inner section **131B**. When the steering cam **40** is in its neutral position, the follower link **128** resides in a juncture **132** situated between the dwell portion **130** and the cam portion **131**.

(91) The operation of the speed assembly **21** will now be described with respect to a steering cam **40** and a speed cam **112** positioned on the right side of the vehicle **10** (as shown, for example, in FIG. 4), to illustrate how the steering input from the steering input device **24** and the speed input from the speed input device **28** may be integrated. FIGS. 14A-14C and 15A-15C schematically show various positions of the speed cam **112**, the follower **120** as controlled by the steering cam **40** (removed for clarity), and the pintle link **102** for different speed and turn combinations for the vehicle **10**.

(92) FIGS. 14A-14C depict a “straight ahead” mode of operation where there is no steering input to the steering input device **24**. FIG. 14A shows a neutral condition where there is no speed input, or the speed input device **28** (FIG. 10) is in the neutral position N. When the driver depresses the speed input device **28** in the first or forward direction, the speed cam **112** is rotated via the speed input shaft **110** (FIG. 10) about pivot **118**. A result of such rotation is depicted in FIG. 14B. This action results in the pintle links **102** being shifted away from the neutral position N, which causes the vehicle **10** to drive in the forward direction. During this process, the steering cam **40** (FIG. 11) remains in a constant default position, which causes the followers **120** to remain at one end of the speed slot **119**. In the illustrated embodiment, this is the bottom end of the speed slot **119**. As shown in FIG. 14C, depressing the speed input device **28** in the second or reverse direction rotates the speed cam **112** in the opposite direction about pivot **118**. Rotation of the speed cam **112** in this opposite direction forces the follower **120** in the opposite direction. This positions the pintle link **102** on the opposite side of the neutral position N, causing the drive unit **29** to drive in reverse.

(93) Operation of the vehicle **10** will now be explained when a turn is directed by the steering input device **24**. Returning to FIGS. 4 and 12, rotating the steering cam **40** in a first direction (e.g., commanding a turn that places the illustrated input member **40** on the outboard side of a turn) causes the follower link **128** to track along the curvature of the inner section **130B** of the dwell portion **130** of the steering slot **127**. The contour of the inner section **130B** is such that the follower link **128** slides in steering slot **127** such that the steering command arm **124** remains stationary and

does not move about pivot **126**. When stationary, the steering command arm **124** does not change the position of the follower **120** in the slot **119** of the speed cam **112**. However, if an extreme turn is intended, such as one that would turn the front wheels about 60 degrees or greater, the steering cam **40** is rotated such that the follower link **128** reaches the end section **130A**. The end section **130A** is contoured so as to cam the follower link **128** and cause the steering control arm **124** to pivot, thereby repositioning the follower **120** to slow the outside transmission drive unit **29** for the extreme turn, as described below.

(94) Alternately, rotating the steering cam **40** counter-clockwise (e.g., commanding a right turn that places the input member **40** on the inboard side of the turn) causes the follower link **128** to move along the curvature of the cam portion **131** of the steering slot **127**. The contour of the inner section portion **131B** is such that the steering cam **40** exerts a force on the follower link **128** causing the steering command arm **124** to move about pivot **126**. As the steering command arm **124** pivots, it moves the follower **120** along the length of the speed slot **119** of the speed cam **112**. This provides a steering input from the steering cam **40** to be integrated with the speed input. That integration produces a “blended output” that is transmitted through the drive rod **104** to the transmission system as a result of an operator manipulating speed input device **28**. A blended output in this context is one that results from a combination of a speed input (e.g., depressing a pedal) and a steering input (e.g., turning a steering wheel). Neither the output from the drive multiplier **116** that travels through drive linkage **38** to drive transmission **30** nor the output from the steer multiplier **112** that travels through steering linkage **48** to steer transmission **32** in U.S. Pat. No. 6,904,985 is a blended output.

(95) Referring now to FIGS. **15A-15C**, FIG. **15A** shows positions of the speed cams **112**, the follower **120** as controlled by the steering cam **40**, and the pintle link **102** in the condition in which the steering input device **24** (FIG. **1**) is rotated to command a maximum inside turn, such that the illustrated speed cam **112** controls the drive unit **29** on the inboard side of the turn. During an inward turn, the steering cam **40** causes the follower **120** to shift in the speed slot **119** toward the opposite end of speed slot **119** from that shown in FIGS. **14A-14C**. Accordingly, when the speed input device **28** is depressed in the first or forward direction as depicted in FIG. **15B**, the geometry of the speed cam **112** for the inward drive unit **29** causes movement of pintle link **102** in a reverse direction. Depressing the speed input device **28** to drive the vehicle forward, with the steering input device **24** fully turned to cause an inward turn, causes pintle link **102** to drive the drive wheel **16** on the inside of the turn in reverse. The follower **120** in the opposing speed cam **112** (not shown) for the outside drive unit **29** does not move toward the upper end of the speed slot **119**. Therefore, the outside wheel is driven forward, resulting in a low- to zero-radius turn.

(96) When the speed input device **28** is depressed in the second or reverse direction, the speed cam **112** rotates in the second direction as depicted in FIG. **15C**. This causes the pintle link **102** to command the inward drive unit **29** to drive the inward drive wheel **16** in the forward direction. Thus, ZTR steering (or at least small-turn radius steering) in forward and reverse is accomplished as a result of the drive units receiving two blended outputs. While the front steerable wheels **18** may rotate in the Ackermann geometry as set forth above, the steering system **20** may be configured to steer the front wheels **18** in any desired manner using sound engineering judgment.

(97) As FIGS. **14A-14C** and **15A-15C** show, the position of the follower **120** within the speed slot **119** may be adjusted by applying a force with the steering cam **40** (as seen in FIG. **11**). Preferably, a bias force, which may be applied by a spring (not shown) coupled to the follower **120** in a manner well known in the art, biases the follower **120** to the neutral position. As the steering input device **24** is turned, the drive rod **104** selectively moves through the speed slot **119** to cross from a first direction position to a second direction position. Preferably, the follower **120** slides in an analog fashion from the bottom to the top of the speed slot **119** depending on the magnitude of the turn directed by the steering input device **24**, establishing a series or a plurality of trajectories through which the follower **120** is selectively maneuvered. Therefore, the follower **120** is selectively

positioned at various points between the first and second maximum positions in the speed slot **119**. In this way, and because the steering cams **40** are rotated independently or asynchronously, the pintle links **102** may be independently controlled through receipt of independent blended outputs from the drive rods **104** to steer and propel the vehicle **10** in a manner consistent with proper steering in the forward and reverse directions. Additionally, the steering cams **40** and the speed cams **112** are preferably configured so that the maximum distance from the neutral position N that the pintle link **102** can be shifted by the follower **120** is greater in the forward direction than in the reverse direction. As a result, a given drive unit **29** (and, more generally, the transmission system) produces a greater maximum magnitude of speed in the forward direction than in the reverse direction. For example, in one embodiment, the vehicle has a maximum forward speed of about 6 mph and a maximum reverse speed of about 4 mph.

(98) Preferably, the steering characteristics of the drive wheels **16** and the front wheels **18** are matched so that the steering provided by the drive wheels **16** and the front wheels **18** cooperate to steer the vehicle **10**. Accordingly, the degree of turn caused by the drive wheels **16** may be matched with the steering angle of the front wheels **18** so that the drive wheels **16** do not try to turn the vehicle in a sharper turn than the front wheels **18**, and vice-versa. In the illustrated embodiment, this is accomplished by selecting the curvature of the steering slot **127** of the steering cam **140** to match the steering angle of the front wheels. This can also reduce the amount of torque required of the drive wheels **16** to turn the vehicle as compared to the amount of torque needed to turn the front castor wheels of some conventional vehicles. With steerable wheels **18**, the operator of the vehicle does not need the level of proficiency required to operate existing lever-controlled ZTR vehicles, and the tendency to damage the driving surface such as by tearing up the grass by skidding the inboard drive wheel during a turn is reduced, and possibly eliminated.

(99) In operation, the steering assembly **20**, via the steering cam **40** on the inboard side of the intended turn, provides a steering input that changes the condition of the speed command to the drive unit **29** received from the speed cam **112** through the assembly **101**. The steering cam **40** on the outboard side of the intended turn does not change the condition of the speed command to the drive unit **29** for small turns.

(100) Speed Curves

(101) For extreme turns, it is preferable for the drive unit **29** on the outboard side to slow so that the front wheels do not plow. FIG. **16** illustrates one example of the wheel speed for the drive wheels **16** produced by the two transmission drive units **29** as a function of steering input, assuming a constant steering input from the speed input device **28** ("constant pedal"). The graph shows that the inside wheel slows more, and more quickly than the outside wheel, during a turn. The inside wheel has a zero speed for a turn of about 90 degrees and has the maximum reverse speed where the inside wheel is turned about 108 degrees. The outside wheel desirably maintains or even slightly increases its speed for turns up to about 60 degrees. The outside wheel gradually slows for larger turns until it slows to a speed of equal magnitude, but in the forward direction, as the inside wheel at 108 degrees to produce a zero turn radius. The FIG. **16** graph of wheel speed vs. applied steer is only one example of how the steering assembly **20**, the speed control assembly **21** and integration device **27** may operate. They may be configured to produce other speed profiles.

(102) The steering assembly **20**, the speed control assembly **21** and integration device **27** work together to provide a reduced average velocity as the vehicle **10** turns, as shown by the FIG. **16** speed curves. The steering assembly **20**, the speed control assembly **21** and integration device **27** work together to balance the torque delivered by the drive wheels **16** and provide the vehicle **10** with infinite and controlled speed modulation through the desired speed ranges of the two transmission drive units **29** from the forward to reverse directions.

(103) A turn results in a steering input to the inward follower **120** that causes the follower **120** to be positioned in the speed slot **119** nearer the point **118** about which the speed cam **112** pivots. This causes the magnitude of the movement of the drive rod **104** to diminish. Correspondingly, the

lateral displacement of the pintle link **102** on the inward side is reduced and the inward drive wheel **16** is driven more slowly. The difference in rotational speed between the drive wheels **16** causes the vehicle **10** to turn. This turn is maintained regardless of the position of the speed cam **112** as long as the setting of the steering input device **24** is not changed. Even as the driver places the vehicle **10** in reverse by switching input on the speed input device **28**, the magnitude of speed on the inward wheel **16** remains smaller than that of the outboard wheel **16**, so that the vehicle continues the turn in the same direction. Thus, consistent or proper steering is maintained when traveling in reverse. Additionally, movement of the steering cams **40** does not reposition the speed cams **112**; it only changes the position at which each follower **120** is positioned in the speed slot **119** of one of the speed cams **112**. And because the speed slot can be configured as an arc having a radius as described above, movement of the steering input device **24** (FIG. 1) does not cause any rotation of the drive wheels **16** or movement of the vehicle **10**. This should accord with the expectation of the operator of the vehicle **10**, who may be accustomed to controlling the movement and speed of the vehicle with one control (e.g., the speed input device **28**) and steering with another control (e.g., the steering input device **24**).

Worm Embodiment

(104) Referring now to FIGS. **17-20**, an alternate embodiment for integrating the steering input from the steering input device **24** and the speed input from the speed input device **28** is illustrated. As in the embodiment above, the drive units (not shown) are coupled to a linkage assembly which includes a pair of drive rods **104A** pivotally coupled to pintle links (not shown). This embodiment illustrates the drive rod **104A** as having a bell crank **149** disposed at one end (and which can be coupled to a bell crank disposed on another drive rod (not shown)) to accommodate the orientation of the transmission drive unit.

(105) FIG. **17** shows the speed input shaft **110** coupled to two speed cams **112A** via a second speed shaft **115A**. Rotation of the speed input shaft **110** causes rotation of the second speed shaft **115A**, which in turn rotates the speed cams **112A**. The speed cams **112A** have a substantially similar shape and substantially similar speed slot **119A** as the speed cams **112** described in the previous embodiment. Followers **120A** positioned at the end of the drive rods **104A** are coupled to the speed cams **112A** with a yoke **121A** and pin **122A** slidably received in the slot **119A**. A further description of the speed cams **112A** and followers **120A** is not needed because they are similar to the speed cams **112** and followers **120** of the embodiment described above.

(106) Two steering cams **40A** are coupled to the chassis **14** such that they rotate about pivot **41A** and are coupled to the steering input device **24** (FIG. 1) via a worm gear **150**. The worm gear **150** is positioned at the end of the steering shaft **30** so that the worm gear **150** is rotated in first and second directions as a result of rotation of the steering input device **24**. The worm gear has first and second variable pitch grooves **152**, **153** cut around its outer circumference. The left steering cam **40A** engages the worm gear **150** via a set pin **154**, and the right steering cam **40A** engages the worm gear via set pin **155**. The set pin **154** is received in the first variable pitch groove **152**. Likewise, the set pin **155** is received in the second variable pitch groove **153**. The variable pitch grooves **152**, **153** are configured to cause the set pins **154**, **155** to selectively pivot the steering cams **40A** as the worm **150** is rotated.

(107) FIG. **18** shows that the variable pitch groove **152** has a dwell portion **152A** in which the variable pitch groove **152** has a first contour. The variable pitch groove **152** also has a cam portion **152B** in which the variable pitch groove **152** has a second contour. The first contour is different than the second contour: the cam portion **152B** has a generally spiral configuration while the dwell portion **152A** extends around the circumference of the worm **150** at a uniform height along the body of the worm. In one embodiment, the dwell portion **152A** and the cam portion **152B** each cover about 240 degrees around the circumference of the worm. However, the length of the dwell portion **152A** and cam portion **152B** may be greater or less than this depending on the desired application, and using sound engineering judgment. When the steering cam **40A** is in its neutral

position, the set pin **154** resides in a juncture **156** between the dwell and cam portions **152A**, **152B** of the variable pitch groove **152**. The second variable pitch groove **153** has a similar dwell portion **153A** and cam portion **153B** that meet at a juncture **157**.

(108) FIG. **18** illustrates a condition in which the set pins **154** and **155** are in neutral positions; specifically, the pins are in the junctures **156**, **157** in their respective grooves **152** and **153**.

(109) FIG. **19** illustrates a second condition after which the worm gear **150** has been rotated by rotation of the steering input device **24** (FIG. **1**). In this second condition, the set pin **154** has traveled through the dwell portion **152A** of groove **152** and the set pin **155** has traveled through the cam portion **153B** of groove **153**.

(110) As best seen in the enlarged view of FIG. **20**, a steering command arm **124A** extends from the steering cam **40A**. The steering command arm **124A** is coupled to the linkage assembly **101A** with the slide **133A** and controls the position of the follower **120A** to provide steering input to the speed cams **112A** in substantially the same way that the steering command arm **124** controls the position of the follower **120** in the embodiment described above.

(111) In operation, the worm gear **150** rotates in response to a steering input on the steering input device **24** (FIG. **1**). When the worm gear **150** is rotated counter-clockwise (e.g., when a left turn is intended that would place the input cam **40A** illustrated in FIG. **20** on the outboard side of the turn), the set pin **155** tracks along the curvature of the dwell portion **153A** of the groove **153**. The contour of the dwell portion **153A** is configured such as the set pin **155** tracks along it, the worm **150** does not cause the steering cam **40A** to rotate about pivot **41A**; instead, the steering cam **40A** remains generally stationary. Thus, the steering command arm **124A** does not cause the follower **120A** to reposition in the slot **119A** of the speed cam **112A**.

(112) Alternately, when the worm gear **150** is rotated clockwise (e.g., when a right turn is intended that would place the input cam **40A** on the inboard side of the turn), the set pin **155** tracks in the cam portion **153B** of the groove **153**. The contour of the cam portion **153B** is configured such that the worm gear **150** exerts a force on the set pin **155** that causes the steering cam **40A** to pivot about pivot **41A**. As the steering cam **40A** pivots, the steering command arm **124A** causes the follower **120A** to shift in the slot **119A** of the speed cam **112A**. The steering cam **40A** on the opposite side responds in similar fashion.

(113) In this embodiment, the steering cam **40A** on the outboard side of the intended turn does not change the position of the follower **120A** with respect to the speed cam **112A**. On the other hand, the steering cam **40A** on the inboard side alters the position of the follower **120A**. The worm gear **150** (and, more particularly, the shape of the variable pitch grooves **152**, **153**) may be configured to cause the transmission system generally (and the outside drive unit specifically) to slow during an extreme turn in order to help prevent plowing of the front wheels **18**. Rotation of the speed cam **112A** through operation of the speed input device **28** and operation of the pintle links by the linkage are substantially the same as the operation of those elements in the embodiment described above and illustrated in FIGS. **14A-14C** and **15A-15C**, and thus need not be repeated.

(114) In the embodiments described above, the vehicle includes right and left steering cams (**40** and **40A**), right and left speed cams (**112** and **112A**), and right and left followers (**120** and **120A**). The follower on the right side of the vehicle is coupled to the right transmission drive unit **29** and is controlled by the right side steering mechanism and right side speed cam. The left follower is coupled to the left transmission drive unit **29** and is controlled by the left side steering mechanism and the left side speed cam. Each steering cam influences the position of its respective follower with respect to the relevant speed cam.

(115) Alternately, the vehicle **10** can include a single steering mechanism interacting with a single speed mechanism with a linkage assembly having a single follower with multiple legs that interact with the transmission system generally, and the transmission drive units **29** more specifically. Additionally, the steering mechanism can change the position of the speed mechanism with respect to the follower in other embodiments of the present devices and systems, which is described next.

Rack and Pinion Embodiments

(116) FIGS. **21-25D** illustrate a speed control assembly **21B** and a portion of a steering assembly **20B**. The steering assembly **20B** includes a steering mechanism in the form of a gear wheel **200** that is externally toothed to engage with a gear or drive chain (omitted from the drawings for simplicity) coupled to the steering input device (e.g., steering input **24**, not shown). Movement of the steering input device by the driver thus rotates the gear wheel **200**. The speed control assembly **21B** includes a speed mechanism comprising master and slave toothed racks **202**, **204** that are coupled to the gear wheel **200** such that they turn along with it, but are capable of moving longitudinally relative to it. As shown in FIG. **22**, this coupling is achieved through lugs **206**, **208** projecting from the gear wheel **200** and slidably received in longitudinal slots **210**, **212** of the respective racks **202**, **204**. Other means for providing a directionally positive arrangement may be adopted. For example, both racks may be slidably coupled (e.g., using bearings) to a base plate (not shown). The base plate may be coupled to the mounting plate **219** (discussed below) with side walls (not shown) to enclose and protect the racks.

(117) The speed cam **21B** also comprises a speed control rack **214** that is coupled to, and movable along its longitudinal direction by, a speed input device (e.g., speed input device **28**, not shown). The speed control rack **214** meshes with a speed control pinion **216**. Both the gear wheel **200** and the speed control pinion **216** are journaled on an axle **217** of a mounting pinion **218**. The axle **217** is journaled in a mounting plate **219** such that it can rotate, but its axis is fixed. Although not shown, the mounting plate **219** may be provided with a slot and the speed control rack **214** may be coupled to the mounting plate **219** with a lug projecting from the speed control rack **214** that rides in the slot. The gear wheel **200** has a domed inner region into which the speed control pinion **216** projects. The dome is cut away to enable meshing of the speed control pinion **216** with the speed control rack **214**. The mounting pinion **218** meshes with the slave rack **204** but runs in an un-toothed longitudinal recess **220** in the master rack **202**, so that it does not restrict longitudinal motion of either rack—when the slave rack **204** moves, the mounting pinion **218** freewheels. The speed control pinion **216** meshes with the master rack **202** so that displacement of the speed control rack **214** produces a corresponding displacement of the master rack **202**.

(118) An integration device comprising a follower pinion **224** (one type of follower) meshes with lower regions of both master and slave racks **202**, **204**. The follower pinion **224** is rotatably mounted on a stub axle **225** carried by a “T” shaped lever **130**. The lever **260** is provided with a fulcrum in the form of a spigot **158** movable along a guideway formed as a slot **160** in the mounting plate **219**, and its left and right limbs are coupled to the ratio control levers **144L**, **144R** (which are comparable in function to the pintle links **102** described above) of the transmission drive units **122L**, **122R** (which can be HSTs as described above, or any other suitable transmission system, such as two continuously variable ratio transmissions, as described below). Although the follower pinion **224** is shown to be co-axial with the mounting pinion **218** in some of the drawings, it is able to move away from this position in response to input from the speed input device (not shown).

(119) The racks **202**, **204**, **214** together form a guide path that is rotatable about a fixed axis defined by the axle **217** by means of the steering input device through the gear wheel **200**. The radial position of the follower pinion **224** (the distance of its center from the fixed axis) is unchanged by rotation of the guide path and depends only on the position of the speed control rack **214**. FIG. **24** shows the configuration when the speed input device is at zero or a neutral position and the steering input device is in a “straight ahead” position. The axis of the follower pinion **224** lies on the fixed axis **217**, and correspondingly the lever **130** (omitted from FIGS. **24-25D** for the sake of representational simplicity) is positioned to place both transmission drive units **122L**, **122R** in neutral position. FIG. **25A** shows the configuration where the steering input device remains at zero (the orientation of the master and slave racks **202**, **204** is the same as in the previous drawing) but the speed input device has caused the speed control rack **214** (not seen in these drawings) to be

advanced, and this motion has been transmitted through the speed control pinion **216** to the master rack **202**. Consequently, the follower pinion **224** has been displaced forwardly from the fixed axis **217**. As in previous embodiments, the effect of this forward displacement is to set the two transmission drive units **122L**, **122R** to identical forward ratios, causing the vehicle **10** to move in a straight line. If the speed control setting of FIG. **25A** is maintained, but the driver moves the steering input device to request a right turn, the configuration of FIG. **25B** is reached. The master and slave racks **202**, **204** have turned through ninety degrees. In the process, both master and slave racks **202**, **204** have rotated around the speed control pinion **216**, causing them to move equally and in opposite directions. Consequently, the radial displacement of the follower pinion **224** from the fixed axis **217** is unchanged. The follower pinion **224** is now displaced laterally to produce a right turn.

(120) Still maintaining the same speed control setting, but moving the steering input device **24** to request a left turn, results in the configuration of FIG. **25C**. Again, the radial displacement of the follower pinion **224** is unchanged.

(121) FIG. **25D** shows the configuration when the steering input device is set to zero but the speed control rack is withdrawn to move the follower pinion **224** rearwardly, setting both transmission drive units **122L**, **122R** to identical reverse ratios and causing the vehicle **10** to reverse in a straight line.

(122) It will be apparent that in the master/slave rack embodiment described above, the speed input device determines the radial distance of the follower or followers from the axis about which the guide path rotates. The displacement of the follower produced by moving the steering input device is a function of this radial distance. Rotating the guide path causes the ratio of one transmission drive unit relative to the other to change, whereas moving the follower along the guide path changes both ratios in the same sense.

(123) FIG. **26** illustrates an arrangement which is functionally similar to that of FIGS. **21-25D** but is believed to be more convenient to assemble. The arrangement includes a master rack **402** and a slave rack **404**, but in this embodiment the racks are received and mounted by a two part housing **450**, **452**. The housing and the racks are able to rotate around axis **454**. Mounting pinion **418** is spatially fixed through an integral boss **456**, which is splined into mounting plate **419**. Housing part **450** has an integral collar **458** through which the housing is rotatably mounted on boss **456**. Running through an axial bore in the mounting pinion **418** is an integral shaft **460** of a speed control pinion **416**, the shaft being splined into an upper gear **462** through which speed control is exercised. The upper gear **462** is coupled to the speed input device through an arrangement (not shown) using either a chain or a further toothed rack. Rotation of the housing **450**, **452** and of the racks it mounts is controlled through a steering gear **464** carried upon the housing and coupled to the steering input device through an arrangement (not shown) using either a further gear, a chain or a further toothed rack. A stub axle **425** mounted on a "T" shaped lever **430** (similar to lever **130** described above) projects into an axial bore of follower pinion **424**. The lever **430** is coupled to the transmission system, and more particularly to two drive units, in the manner described above with respect to FIGS. **21-23**. The follower pinion **424** meshes with both master and slave racks **402**, **404**. Speed control pinion **416** meshes only with the master rack **402**, so that moving this pinion by means of the speed input device moves the follower pinion **424** radially. Fixed mounting pinion **418** meshes only with the slave rack **404** to ensure that when the housing rotates, the slave rack retreats to compensate for the advance of the master rack. As a result, rotation of the housing does not in itself change the radial position of the follower pinion **424**.

(124) Assembly of this arrangement involves placing all of the relevant parts in housing part **450**, then adding housing part **452** to keep them in place. Although it is not apparent from the drawing, the housing **450**, **452** forms an elongate enclosure containing the full length of the racks and leaving them room to move longitudinally. Stub axle **425** and a surrounding, projecting hub **464** project through an elongate slot in the housing part **452** to give them freedom to move

longitudinally. Seals, including “O” ring seals **466**, **468**, retain lubricant in the housing **450**, **452**. Mounting the housing assembly on the mounting plate **419** is achieved by inserting the shaft **460** through its hole in the mounting plate and securing the upper gear **462** in place upon the shaft **460** to resist its subsequent withdrawal.

(125) FIGS. **27** and **28** show a version of a transmission arrangement designed to match the characteristics of an Ackerman-type wheel assembly **50**. The mechanism seen at **500** serves to control the position of the T-shaped lever **502**, which is equivalent to the T-shaped lever seen in FIGS. **21-23**. In this embodiment, the outer ends of this lever couple to the ratio control levers of the variators (which are not seen in this drawing) through spherical heads **503** received in complementarily shaped slots **504**, which is a slight modification of the FIGS. **21-23** embodiment. A more significant difference of the present arrangement concerns an arrangement of gears **506**, **508**, through which the mechanism **500** is coupled to the steering input device (not shown). The gear wheel **506** serves the same purpose as gear wheel **200** seen in FIGS. **21-23**: it serves to rotate the mechanism **500** by turning the lever **502** to provide the required steering effect. The driver is able to turn the gear wheel **506** through the steering input device (e.g., steering input device **24** from FIG. **1**), which is coupled to the steering gear **508** that meshes with the gear wheel **506**. The gear wheel **506** and the steering gear **508** are non-circular, and their shapes are chosen to provide the required relationship between the position of the steering input device and the ratios provided by the two transmission drive units (e.g., drive units **29** or **122L**, **122R** described above).

Determining the shapes for the two gears **506**, **508** is a straightforward numerical exercise based upon the characteristic (steering input device position vs. vehicle turn radius) of the Ackermann steering device and the characteristic (ratio control lever position vs. ratio) of the transmission drive units. In the present embodiment, this yields a shape for the gear wheel **506** that has three curved sides, as seen. The gears **506**, **508** are shaped to remain in mesh at all times, so that the shape of one determines the shape of the other.

(126) FIGS. **29-31** depict the construction of a continuously variable ratio transmission (CVT) having a geared neutral condition that is suitable for use as a transmission drive unit **29**. The depicted drive unit is a toroidal-race, rolling-traction type, although other types of CVTs may be used. For example, a “belt and sheave” type transmission that could be used consistently with the present systems and vehicles is disclosed in U.S. Pat. No. 5,766,105, which is incorporated by reference.

(127) The illustrated CVT comprises a variator **V** having a toroidally-recessed input disc **310** and a facing toroidally-recessed output disc **312**. Two rollers **314**, **316** are mounted in the toroidal cavity defined between the opposing toroidally-recessed faces of the input and output discs **310**, **312** to transmit drive from the input disc **310** to the output disc **312** with a ratio that can be varied by tilting the rollers **314**, **316**.

(128) The input disc **310** is coupled to, and rotates with, a transmission input shaft **318** which is driven from the vehicle's engine (e.g., engine **12** of vehicle **10**). The variator **V** provides an output via a tubular output shaft **320** which is coupled to the output disc **312** and arranged coaxially with, and around, the input shaft **318**. The input shaft **318** and the variator output shaft **320** provide the inputs to a compound mixing epicyclic gear train **E1**. As shown schematically, the end of the variator output shaft **320** remote from the output disc **312** carries a first sun gear **S1** of the mixing epicyclic gear train **E1**. The carrier **C1** of the gear train **E1** is coupled to, and driven by, the input shaft **318**. The carrier **C1** carries four identical equally-spaced radially inner planet gears **P1** and four identical equally-spaced radially outer planet gears **P2** of the same size as the radially inner planet gears **P1**. The radially inner planet gears **P1** engage with the first sun gear **S1** and with a respective one of the four radially outer planet gears **P2**. The radially outer planet gears **P2** also engage with an internally-toothed annulus **A1**, which forms the output of the mixing epicyclic gear train **E1**. The output from the annulus **A1** is coupled via tubular coaxial output shaft **322** to a simple reducing epicyclic gearset **E2**. The reducing epicyclic gearset **E2** comprises an input sun gear **S2**

carried by shaft **322** which meshes with four equally angularly spaced planet gears **P3** carried by carrier **C2**. The planet gears **P3** also mesh with an annulus **A2** fixed to the transmission housing. The rotation of the carrier **C2** forms the output of the reducing epicyclic gear set **E2** and is transmitted to the exterior by an output shaft **24** which is coupled to the carrier **C2**. The output shaft **324** is coaxial with the input shaft **318**, one end of which is received in a recess **326** in the innermost end of the output shaft **324**. The output shaft **324** is coupled to the relevant driven vehicle wheel.

(129) The transmission is housed in a generally tubular casing **330** which supports the input and output shafts **318**, **320**. The end of the casing **330** adjacent the input shaft **318** is closed off by means of an end plate **332**. A conical Belleville spring washer **334** extends between the inner face of the end plate **332** and an annular bearing plate **336**, which is in rolling contact with an outer planar face of the variator input disc **310**. The Belleville spring washer applies a force (an “end load”) to the input disc **310** and permits torque to be transmitted from the input disc **310** via the rollers **314**, **316** to the output disc **312**.

(130) By varying the inclination of the two rollers **314**, **316** (as described below), the speed of the output disc **312** relative to the input disc **310** can be varied. By combining the rotations of the transmission input and variator output in the mixing epicyclic gear train **E1**, the output of the transmission can be varied. In the arrangement illustrated, the transmission can be varied between full reverse, through “geared neutral” to full forward, as well as anywhere in between. However, the operating range of the variator can be tailored to requirements by appropriate selection of the gearing. For example, the variator may be arranged to vary between low reverse through geared neutral to high forward overdrive if a vehicle to which the transmission were fixed operated normally in forward gear and operated only occasionally in reverse.

(131) The mechanism for varying the inclination of the two rollers **314**, **316** is shown in more detail in FIG. **30**. Each roller **314**, **316** is rotatably mounted in a roller carriage **340** by means of a stub axle **342** which is rotatably mounted in opposed planar support plates **44**, **46** of the roller carriage. One end of each of the roller carriages **340** is coupled to a respective one of the two ends of the cross-bar **348** of a control lever **350** by means of a spherical bearing **352** (e.g., “Rose bearing” manufactured by Rose Bearings Limited). The control lever **348** is provided with a pivot pin **354** located mid-way between the center points of the two spherical bearings **352**. The pivot pin is received in a slot **356** of the same width as the diameter of the pivot pin but elongated in the radial direction with respect to the rotational axis of the variator. The slot **356** is provided in a mounting lug **358** which projects into the variator into the space between the input and output discs **310**, **312**.

(132) The lever **350** is provided with an actuating arm **360** which projects out the variator housing in a direction perpendicular to the line joining the center points of the two spherical bearings **352** (perpendicular to the axis of the cross-bar **348** of the lever). This arm **360** forms the lever through which the transmission ratio is controlled and corresponds to the ratio control levers **144L**, **144R** described in connection with FIGS. **22-25E**. As the lever **350** pivots, one of the rollers **310**, **312** is pushed and the other is pulled, both with equal torque. The mounting of the pivot pin **354** within the slot **356** in the mounting lug **358** allows the pin **354** to move radially inwardly and outwardly, which ensures that the horizontal forces from the rollers are equalized and cancel each other out. This may be valuable with low-cost assemblies, where the manufacture of the components is likely to be less accurate. The radial movement of the pivot of the lever allows the lever to move to a position in which any imbalance between the two rollers arising from manufacturing differences will be cancelled out.

(133) It will be apparent that when drive is transmitted, the rollers are subject to a net torque tending to drive them circumferentially about the variator axis. This torque must be reacted to a fixed point for the rollers to hold steady positions. The necessary reaction torque is provided by the lever **360**, so that the force upon the lever is related to the torques at the transmission input and output. When, for example, one wheel tends to lag behind the vehicle speed, in a way that could

otherwise cause it to slip, the effect is to change the force upon the lever such that the speed of the relevant wheel tends to increase. By permitting this adjustment, the depicted arrangements reduce or even eliminate wheel slip.

(134) Descriptions of well known manufacturing and assembly techniques, components and equipment have been omitted so as not to unnecessarily obscure the present systems and devices in unnecessary detail. The present systems and devices are not intended to be limited to the particular forms disclosed. Rather, they are to cover all modifications, equivalents, and alternatives falling within the scope of the claims.

(135) For example, the steering assembly that receives a steering input from the steering input device may be configured differently than shown in the figures. In alternative embodiments, the steering mechanism for a given vehicle may be a single steering cam with two steering slots, rather than two steering cams with one steering slot each, as shown for example in FIG. 12. Furthermore, such a dual-slotted steering cam may be oriented horizontally (or generally perpendicular with the ground), instead of being oriented vertically like the steering cams shown in the figures. Moreover, such a steering cam (like any of the present steering cams) may be canted at any angle suited to a given application and chosen using sound engineering judgment.

(136) Another alternative includes moving the gear set that initially translates the rotation of a steering input device (such as a steering wheel) into movement that is transmitted to the wheel assemblies. For example, such a gear set could be moved forward and positioned in between two rods that otherwise act as tie rods linking the two front wheel gear assemblies together.

(137) As yet another example, the steering slots that are shown in the figures as positioned in the steering cams could be instead positioned in one of the gears making up the gear assemblies for the front steerable wheels.

(138) As still another example, the vertically-oriented speed cams could be made to mesh with each other to a certain degree and oriented horizontally.

(139) The appended claims are not to be interpreted as including means-plus-function limitations, unless such a limitation is explicitly recited in a given claim using the phrase(s) “means for” and/or “step for,” respectively.

Claims

1. A vehicle comprising: a chassis; a left front steerable, non-driving wheel coupled to the chassis and capable of turning more than 90 degrees left; a right front steerable, non-driving wheel coupled to the chassis and capable of turning more than 90 degrees right; left and right rear drivable wheels coupled to left and right hydrostatic drive units; a steering wheel that can receive a steering input from an operator of the vehicle; a steering assembly coupling the steering wheel to the left front and right front steerable, non-driving wheels such that they will follow substantially-true Ackermann steering in response to turning the steering wheel to cause a zero radius turn, the steering assembly including: a first gear pair for turning the right front steerable, non-driving wheel, the first gear pair comprising: a first wheel gear having a first wheel gear pivot axis and a first wheel gear pitch line that changes in distance from the first wheel gear pivot axis; and a first steering gear engaged with the first wheel gear, the first steering gear having a first steering gear pivot axis and a first steering gear pitch line that changes in distance from the first steering gear pivot axis; a second gear pair for turning the left front steerable, non-driving wheel, the second gear pair comprising: a second wheel gear having a second wheel gear pivot axis and a second wheel gear pitch line that changes in distance from the second wheel gear pivot axis; and a second steering gear engaged with the second wheel gear, the second steering gear having a second steering gear pivot axis and a second steering gear pitch line that changes in distance from the second steering gear pivot axis; a speed pedal that can receive a speed input from an operator of the vehicle; first and second speed cams oriented parallel to each other and coupled to the speed pedal,

the first speed cam including a first curved slot and the second speed cam including a second curved slot, the first and second curved slots having the same shape, where movement of the speed pedal causes the first and second speed cams to rotate in the same direction and by the same amount about an axis extending through each of the first and second curved slots; a first follower that rides in the first curved slot and that will move in response to steering wheel movement to a right-hand turn from a straight-ahead position; and a second follower that rides in the second curved slot and that will move in response to steering wheel movement to a left-hand turn from a straight-ahead position; where the left and right hydrostatic drive units are capable of driving the left and right rear drivable wheels at different speeds and in different directions; movement of the steering wheel without a speed input from the speed pedal does not cause rotation of the left and right rear drivable wheels; and the same one of the left and right rear drivable wheels will rotate faster than the other of the left and right rear drivable wheels in both forward and reverse for a turned position of the steering wheel.

2. The vehicle of claim 1, further comprising: an engine mounted to the chassis.

3. The vehicle of claim 2, further comprising: an operator station supported by the chassis, the operating station comprising a seat.

4. The vehicle of claim 3, further comprising: a mower deck positioned rearward of the left and right front steerable, non-driving wheels.

5. The vehicle of claim 4, where: the left and right rear drivable wheels each has a fixed rotational axis.

6. The vehicle of claim 2, further comprising: a mower deck positioned rearward of the left and right front steerable, non-driving wheels.

7. The vehicle of claim 6, where: the left and right rear drivable wheels each has a fixed rotational axis.

8. The vehicle of claim 1, further comprising: an operator station supported by the chassis, the operating station comprising a seat.

9. The vehicle of claim 1, further comprising: a mower deck positioned rearward of the left and right front steerable, non-driving wheels.

10. The vehicle of claim 1, where: the left and right rear drivable wheels each has a fixed rotational axis.

11. A vehicle capable of making a small radius turn, comprising: a chassis; a left front steerable, non-driving wheel coupled to the chassis; a right front steerable, non-driving wheel coupled to the chassis, the left and right front steerable, non-driving wheels rotatable to a zero radius turn; left and right rear drivable wheels coupled to left and right hydrostatic drive units, the left and right rear drivable wheels each having a fixed rotational axis; a steering wheel coupled to a steering pinion, the steering pinion coupled to the right front steerable, non-driving wheel; a steering assembly coupling the steering wheel to the left front and right front steerable, non-driving wheels such that they will follow substantially-true Ackermann steering in response to turning the steering wheel to cause the zero radius turn, the steering assembly including: a first gear pair for turning the right front steerable, non-driving wheel, the first gear pair comprising: a first wheel gear having a first wheel gear pivot axis and a first wheel gear pitch line that changes in distance from the first wheel gear pivot axis; and a first steering gear engaged with the first wheel gear, the first steering gear having a first steering gear pivot axis and a first steering gear pitch line that changes in distance from the first steering gear pivot axis; a second gear pair for turning the left front steerable, non-driving wheel, the second gear pair comprising: a second wheel gear having a second wheel gear pivot axis and a second wheel gear pitch line that changes in distance from the second wheel gear pivot axis; and a second steering gear engaged with the second wheel gear, the second steering gear having a second steering gear pivot axis and a second steering gear pitch line that changes in distance from the second steering gear pivot axis; a speed pedal that can receive a speed input from an operator, the speed pedal biased to a neutral position and movable to forward and reverse

positions; first and second speed cams oriented parallel to each other and coupled to the speed pedal, the first speed cam including a first curved slot and the second speed cam including a second curved slot, the first and second curved slots having the same shape, where movement of the speed pedal causes the first and second speed cams to rotate in the same direction and by the same amount about an axis extending through each of the first and second curved slots; a first follower comprising a first pin positioned in the first curved slot, where the first pin has a position in the first curved slot from which the first pin will move in response to steering wheel movement to a right-hand turn from a straight-ahead position, the first follower being coupled to the right hydrostatic drive unit; and a second follower comprising a second pin positioned in the second curved slot, where the second pin has a position in the second curved slot from which the second pin will move in response to steering wheel movement to a left-hand turn from a straight-ahead position, the second follower being coupled to the left hydrostatic drive unit; where the left and right hydrostatic drive units are capable of driving the left and right rear drivable wheels at different speeds and in different directions; the vehicle has a greater maximum forward speed than maximum reverse speed; movement of the steering wheel without a speed input from the speed pedal does not cause rotation of the left and right rear drivable wheels; and the same one of the left and right rear drivable wheels will rotate faster than the other of the left and right rear drivable wheels in both forward and reverse for a turned position of the steering wheel.

12. The vehicle of claim 11, further comprising: an engine mounted to the chassis.

13. The vehicle of claim 11, further comprising: an operator station supported by the chassis, the operating station comprising a seat.

14. The vehicle of claim 13, further comprising: a mower deck positioned rearward of the left and right front steerable, non-driving wheels.

15. The vehicle of claim 11, further comprising: a mower deck positioned rearward of the left and right front steerable, non-driving wheels.

16. A vehicle capable of making a small radius turn, comprising: a chassis; a left front steerable, non-driving wheel coupled to the chassis; a right front steerable, non-driving wheel coupled to the chassis; left and right rear drivable wheels coupled to left and right hydrostatic drive units, the left and right rear drivable wheels each having a fixed rotational axis; a steering wheel coupled to a steering pinion, the steering pinion coupled to the right front steerable, non-driving wheel; a steering assembly coupling the steering wheel to the left front and right front steerable, non-driving wheels such that they will follow substantially-true Ackermann steering in response to turning of the steering wheel to achieve a zero-radius turn, in either direction, the steering assembly including: a first gear pair for turning the right front steerable, non-driving wheel, the first gear pair comprising: a first wheel gear having a first wheel gear pivot axis and a first wheel gear pitch line that changes in distance from the first wheel gear pivot axis; and a first steering gear engaged with the first wheel gear, the first steering gear having a first steering gear pivot axis and a first steering gear pitch line that changes in distance from the first steering gear pivot axis; a second gear pair for turning the left front steerable, non-driving wheel, the second gear pair comprising: a second wheel gear having a second wheel gear pivot axis and a second wheel gear pitch line that changes in distance from the second wheel gear pivot axis; and a second steering gear engaged with the second wheel gear, the second steering gear having a second steering gear pivot axis and a second steering gear pitch line that changes in distance from the second steering gear pivot axis; a speed pedal that can receive a speed input from an operator, the speed pedal biased to a neutral position and movable to forward and reverse positions; first and second speed cams oriented parallel to each other and coupled to the speed pedal, the first speed cam including a first curved slot and the second speed cam including a second curved slot, the first and second curved slots having the same shape, where movement of the speed pedal causes the first and second speed cams to rotate in the same direction and by the same amount about an axis extending through each of the first and second curved slots; a first yoke having a first pin positioned in the first curved slot, where the first

pin has a position in the first curved slot from which the first pin will move in response to steering wheel movement to a right-hand turn from a straight-ahead position, the first yoke being coupled to the right hydrostatic drive unit through one or more rods; and a second yoke having a second pin positioned in the second curved slot, where the second pin can has a position in the second curved slot from which the second pin will move in response to steering wheel movement to a left-hand turn from a straight-ahead position, the second yoke being coupled to the left hydrostatic drive unit through one or more rods; where the left and right hydrostatic drive units are capable of driving the left and right rear drivable wheels at different speeds and in different directions; the vehicle has a greater maximum forward speed than maximum reverse speed; movement of the steering wheel without a speed input from the speed pedal does not cause rotation of the left and right rear drivable wheels; and the same one of the left and right rear drivable wheels will rotate faster than the other of the left and right rear drivable wheels in both forward and reverse for a turned position of the steering wheel.

17. The vehicle of claim 16, further comprising: an engine mounted to the chassis.

18. The vehicle of claim 16, further comprising: an operator station supported by the chassis, the operating station comprising a seat.

19. The vehicle of claim 18, further comprising: a mower deck positioned rearward of the left and right front steerable, non-driving wheels.
